



HACKATHON

GCC Edition

[illegible]

Problem Brief

TARGET PERSONAS

OUR SOLUTION

Solution Overview

End-to-end governed conversational analytics on Snowflake, built entirely with CoCo

WHAT WE BUILT

TOTAL: 52+ Snowflake objects deployed

Architecture Diagram

Data flow from source tables through governed semantic layer to multi-agent conversational interface

Data Foundation & Pipeline Creation

Synthetic data generation + INCREMENTAL Dynamic Tables + Stream-triggered alerts

SYNTHETIC DATA GENERATION (built entirely with CoCo)

INCREMENTAL DYNAMIC TABLES

PIPELINE: Tables → DTs (incremental) → View (live risk) → Stream → Task → Alerts

```
#include <algorithm>
#include <cstring>

using namespace std;

const int N = 100000;
const int debug_mode = false;

//int main() {
//    int n, m;
//    cin >> n >> m;
//    int a[N + N], delta[N];
//    for (int i = 1; i <= n; i++) {
//        cin >> a[i];
//        delta[i] = a[i] - a[i - 1];
//    }
//    for (int i = 1; i <= m; i++) {
//        int l, r;
//        cin >> l >> r;
//        int ans = 0;
//        for (int j = l; j <= r; j++) {
//            ans += delta[j];
//        }
//        cout << ans << " ";
//    }
//    return 0;
//}
```

YOUR STORY



H2S

Semantic Model & Ontology Authoring

Single source of truth: 4 canonical metrics enforced across all personas

SEMANTIC VIEW: SUPPLY-CHAIN ONTOLOGY

- Built with CoCo using `SYSTEM$CREATE_SEMANTIC_VIEW_FROM_YAML`
- References 4 Dynamic Tables + 1 View as logical tables
- Cortex Analyst generates SQL from natural language questions

CoCo CLI

4 CANONICAL METRICS (consistent across all personas)

OTD% = $\text{on_time_count} / \text{total_shipments} \times 100$ (delivery performance)

Fill Rate% = $\text{fulfilled_qty} / \text{ordered_qty} \times 100$ (order fulfillment)

DOI = $\text{on_hand_qty} / \text{avg_daily_demand}$ (inventory coverage)

Landed Cost = $\text{material} + \text{freight_per_unit} + \text{duty_per_unit}$ (total cost)

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VERIFIED QUERIES

- Pre-validated SQL for common questions per persona
- Verified at epoch timestamp for audit trail
- Agent always returns the same answer for the same question — no metric drift

THANK YOU

Cross-persona consistency: Procurement, Planning, and Logistics see identical numbers

Multi-Agent Orchestration

4 Cortex Agents with intelligent routing, shared semantic view, and domain skills

AGENT ARCHITECTURE

SUPPLY_CHAIN_AGENT (Primary Router)

PROCUREMENT_AGENT

PLANNING_AGENT

LOGISTICS_AGENT

Custom Tools & Guardrails

5 registered tools with input validation, confirmation gates, and governed write paths

CUSTOM TOOLS (type: generic + input_schema + execution_environment)

GUARDRAIL CHAIN (every write action)

MCP Connectors: Jira & Gmail

Agents read from and act across external tools — not just text responses

ATLASSIAN MCP (SUPPLY_CHAIN_DB.SCM.ATLASSIAN_MCP)

EXPEDITE WORKFLOW (end-to-end with MCP)

EMAIL MCP (SUPPLY_CHAIN_DB.SCM.EMAIL)

Clean separation: SPs handle guardrails + DB writes, Agents handle MCP calls

CoCo Automations & Scheduled Runs

3 unattended daily briefs with Gmail delivery — the solution operates on autopilot

3 TEAM-BASED DAILY BRIEFS

LOGISTICS DAILY BRIEF — Weekdays 7:30 AM IST

PLANNING DAILY BRIEF — Weekdays 8:00 AM IST

PROCUREMENT DAILY BRIEF — Weekdays 8:30 AM IST

DELIVERY & MONITORING

Streamlit Dashboard & Reusable Skills

Interactive monitoring app + 5 published skills for agent specialization

STREAMLIT DASHBOARD (deployed on Snowflake)

5 REUSABLE SKILLS (YAML frontmatter, stage-hosted)

Data Governance & RBAC

Column masking, role hierarchy, and agent-safe access controls

5-ROLE HIERARCHY

COLUMN MASKING POLICIES

AGENT GUARDRAILS (governed behavior)

Impact Statement

Measurable outcomes, scalability, and real-world deployment path

MEASURABLE OUTCOMES

SCALABILITY

BEYOND THE DEMO

Solution Completeness: 52+ Objects

Every CoCo capability demonstrated — from data generation to cross-tool action

Deployed
SNOWFLAKE OBJECTS

CoCo FEATURES DEMONSTRATED

Thank You

Supply Chain Ontology & Governed Conversational Analytics

Prabhjot Kaur • Solo Team • GCC Edition

52+ Snowflake objects • 4 Cortex Agents • 2 MCP Servers
5 Reusable Skills • 3 Automations • 1 Semantic Ontology